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RA-25-0116

April 21, 2025

10 CFR 50.73

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
11555 Rockville Pike
Rockville, MD 20852-2746

Duke Energy Carolinas, LLC
Oconee Nuclear Station Units 1, 2, and 3
Docket Numbers: 50-269, 50-270, and 50-287
Subsequent Renewed Operating Licenses: DPR-38, DPR-47, and DPR-55

Subject: Licensee Event Report 269/2025-001, Revision 00 – Inoperability of Keowee Hydroelectric Unit Due to Legacy Deficiencies in Generator Exciter Speed Switch Calibration Procedure Resulting in a Condition Prohibited by Technical Specifications

Licensee Event Report 269/2025-001, Revision 00, is being submitted pursuant to the requirements of 10 CFR 50.73 to provide notification of the subject event.

There are no regulatory commitments associated with this LER.

There are no unresolved corrective actions necessary to restore compliance with NRC requirements.

If there are questions, or further information is needed, contact David Wilson, Regulatory Affairs, at (864) 873-3451.

Sincerely,

A handwritten signature in black ink, appearing to read "Steve Snider", written in a cursive style.

Steven M. Snider
Vice President
Oconee Nuclear Station

Enclosure: Licensee Event Report 269/2025-001 Rev.00

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cc (w/Enclosure):

Mr. Julio Lara, Administrator (Acting), Region II
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Mr. Shawn Williams, Project Manager
U.S. Nuclear Regulatory Commission
11555 Rockville Pike
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Rockville, MD 20852-2738

Mr. Nick Smalley
NRC Senior Resident Inspector
Oconee Nuclear Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Oconee Nuclear Station Unit 1	☒ 050 ☐ 052	2. Docket Number 0269	3. Page 1 OF 4
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4. Title Inoperability of Keowee Hydroelectric Unit Due to Legacy Deficiencies in Generator Exciter Speed Switch Calibration Procedure Resulting in a Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
02	20	2025	2025	- 001 -	00	04	21	2025	Oconee Nuclear Station Unit 2 and Oconee Nuclear Station Unit 3		0270, 0287
									Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER**

Licensee Contact David Wilson, Oconee Regulatory Affairs	Phone Number (Include area code) (864) 873-3451
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	EK	SIS	D283	Y					

14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)									

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 2/20/2025, Keowee Hydroelectric Station Unit 1 (KHU-1) was determined to be inoperable based on observed generator excitation time exceeding design basis values. Troubleshooting determined that speed switches responsible for automatically initiating generator excitation during KHU startup were not fully calibrated following planned replacement on 2/18/2025 due to a legacy procedure deficiency and the scope of post-maintenance testing (PMT) was inadequate to detect the incomplete calibration. The switches removed on 2/18/2025 were reinstalled, correct excitation response timing verified, and KHU-1 was restored to operable on 2/21/2025. Planned corrective actions include speed switch calibration procedure revisions to correct the procedure deficiencies and PMT scope, reviews for similar deficiencies in other instrument procedures, and reviews for adequacy of PMT scope for other complex hardware configurable electronic devices.

This event is reportable under 10 CFR 50.73(a)(2)(i)(B) as an Operation or Condition Prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Oconee Nuclear Station Unit 1	☒ 050	2. DOCKET NUMBER 00269	3. LER NUMBER				
	☐ 052		<table border="1"><thead><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr></thead><tbody><tr><td>2025</td><td>- 001</td><td>- 00</td></tr></tbody></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025
YEAR	SEQUENTIAL NUMBER	REV NO.					
2025	- 001	- 00					

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

BACKGROUND

At Oconee Nuclear Station (ONS), the Keowee Hydroelectric Station (KHS) [EK] serves the emergency power function typically performed by diesel generators [DG] at other nuclear facilities. The KHS consists of two (2) hydroelectric turbine/generator units (KHUs) and associated support equipment and auxiliaries. Each KHU is provided with its own automatic start equipment and both KHUs undergo simultaneous automatic emergency starts during an emergency. Emergency power can be provided to the main feeder buses (MFB)[EB] of any or all three ONS Units from the Keowee Hydroelectric Station via either of two emergency power paths:

- The Overhead Emergency Power Path, that includes the Unit's Start-up Transformer [XFMR] (CT-1, CT-2, or CT-3) on each ONS Unit
- The Underground Emergency Power Path, that includes Transformer CT-4 [XFMR] and both Standby Buses [EB].

Emergency onsite power may also be provided to the MFB of any or all three ONS Units from a Lee Combustion Turbine (LCT) [EK] on a dedicated 100kV transmission line through Transformer CT-5 [XFMR] via one or both Standby Buses.

For any ONS Unit in Modes 1-4, Technical Specification (TS) 3.8.1 specifies that two KHUs be operable with one capable of automatically providing power through the Underground Emergency Power Path to both MFB via the Standby Bus and the other capable of automatically providing power through the Overhead Emergency Power Path to both MFB. An operable KHU and its required emergency power path are required to be able to provide sufficient power within specified limits of voltage and frequency within 23 seconds after an emergency start initiate signal.

EVENT DESCRIPTION

All times are Eastern Standard Time.

On 2/18/2025, ONS Units 1, 2, and 3 were all in Mode 1 operating at approximately 100% full power, KHU-1 aligned as the Overhead KHU, and KHU-2 aligned as the Underground KHU. At 08:14, KHU-1 was removed from service for a planned maintenance outage. An item in the scope of the planned maintenance was planned replacement of speed switches in the KHU-1 generator exciter control circuit. These switches initiate generator field energization during KHU startup once the KHU reaches a preset speed. Following completion of maintenance and operational checks, KHU-1 was declared operable at 22:36. An alignment swap of the KHUs was completed at 23:09, resulting in KHU-1 being aligned as the Underground KHU and KHU-2 being aligned as the Overhead KHU.

During the normal operational check of KHU-1 performed on 2/18/2025 following maintenance, operators noted that KHU-1 generator excitation seemed to take longer than usual. However, no unexpected alarms were received. Operators continued to monitor performance of KHU-1 during normal commercial startups over the next few shifts. On 2/20/2025, Engineering was contacted to evaluate the KHU-1 startup performance. Data review indicated that a delay KHU-1 generator excitation ranged from approximately 50 seconds to as much as approximately 2 minutes during normal, non-emergency starts. Engineering confirmed that the excitation logic was the same for normal and emergency start and that, while KHU-1 continued to be capable of starting and supplying emergency power, the observed additional time delay challenged the ability of KHU-1 to meet the requirement to provide emergency power within 23 seconds of receipt of an emergency start signal.

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NARRATIVE**EVENT DESCRIPTION (continued)**

Based on this determination, operations declared KHU-1 inoperable on 2/20/2025 at 21:34. Maintenance replaced the speed switches with those originally installed and, following successful operational checks (including verification of excitation timing), operations declared KHU-1 operable on 2/21/25 at 14:29.

There were no changes to status of ONS Units 1, 2, or 3, KHU-2 remained operable and both LCTs were available during this event. Also, no significant structures, systems, or components were out of service that contributed to the event.

Reportability

Investigation determined that the direct cause of the KHU-1 excitation delay was associated with the speed switches installed during the planned maintenance outage. All requirements were fully implemented for: TS 3.8.1 Condition C during the planned maintenance outage (i.e., 2/18/25 08:14 – 22:36) where KHU-1 was the Overhead KHU and TS 3.8.1 Condition D following determination KHU-1 was inoperable due to the speed switch issue (i.e., 2/20/25 21:34 – 2/21/25 14:29) where KHU-1 was the Underground KHU. However, since KHU-1 was never in fact operable following completion of planned maintenance, TS 3.8.1 Condition D would have retroactively applied beginning 2/18/25 08:14, resulting in TS 3.8.1 required actions not being completed within the TS specified completion times. Therefore, this issue is being reported under 10 CFR 50.73(a)(2)(i)(B) as an Operation or Condition Prohibited by TS.

CAUSAL FACTORS

Maintenance technical procedures for performing speed switch replacement and calibration were inadequate. The procedure contained a legacy issue in which a series of procedure revisions performed in 2010-2012 replaced procedure steps with informational notes. This resulted in incomplete setup of the speed switches, which caused the observed delays in actuation. In addition, the scope of post-maintenance testing guidance specified in the procedure was inadequate to identify a speed switch with an incomplete setup.

CORRECTIVE ACTIONSCompleted

- Reinstalled original speed switches and verified response time was as required.

Planned

- Revise Dynalco speed switch procedures to ensure required actions are in steps and notes do not provide directive actions.
- Perform extent of condition review of other instrument calibration procedures to ensure informational notes do not provide directive actions.
- Review post-maintenance testing strategy for Dynalco speed switches to ensure specified testing will detect potential speed switch setup issues.
- Review post-maintenance testing strategy in instrument procedures for complex hardware configurable electronic devices to ensure configured functions are adequately tested.
- Communicate learnings to Engineering, Maintenance, and Procedures groups regarding ensuring all required scope is included in the technical procedure steps and performed following component replacements.
- Communicate learnings to Operations highlighting timely identification of unexpected issues using the Corrective Action Program.

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Oconee Nuclear Station Unit 1

☒ 050☐ 052**2. DOCKET NUMBER**

00269

3. LER NUMBER

YEAR

2025

SEQUENTIAL
NUMBER

- 001

REV NO.

- 00

NARRATIVE**SAFETY ANALYSIS**

A probabilistic risk assessment (PRA) evaluation was performed for the period the Keowee Hydro Unit 1 was inoperable to determine the significance of this event. It was determined through a quantitative analysis that the event had a very low impact on the incremental conditional core damage probability (ICCDP) and incremental conditional large early release probability (ICLERP). Therefore, it has been determined the condition had an insignificant impact to public health and safety. This was largely due to the sustained availability of the Keowee Hydro Unit 2, Lee Combustion Turbines, Standby Shutdown Facility, and the Protected Service Water system. These systems provided adequate redundancy to ensure capabilities to mitigate accident scenarios were maintained while the Keowee Hydro Unit 1 was inoperable.

ADDITIONAL INFORMATION

A review of Duke Energy's Corrective Action Program identified one related Oconee Licensee Event Report (LER) in the past three years that involved similar underlying cause(s), failure(s), or sequence of events. LER 269/2024-001-00 reported an inoperability of the ONS Unit 1 Standby Shutdown Facility (SSF) Reactor Coolant System (RCS)[AB] Pressurizer [PZR] heaters [EHTR] due to miscalibration of a current switch [IS] in the control circuit. The SSF Pressurizer heater current switch, like the KHU speed switch in the present LER, is a complex hardware configurable electronic device. In both issues, the associated component replacement procedures contained setup information in informational notes rather than in procedure steps, which lead to incomplete setups of the associated devices during planned component replacements. Also, there were inadequacies with post-maintenance testing such that the incomplete component setups were not detected. These procedure deficiencies were legacy in nature (the SSF issue involved procedure revisions in the late 1990s and the KHU issue involved several procedure revisions from 2010-2012) and involved portions of the associated procedures that had not been performed prior to the associated events. Extent of cause actions from the SSF Pressurizer heater issue were focused on component procedures for similar current switches. Based on the similarities between the two events, corrective actions discussed above were developed to perform a review of instrument calibration procedures for complex hardware configurable electronic devices to ensure:

- informational notes do not provide directive actions
- post-maintenance testing scope ensures configured functions are adequately tested.

This event is INPO IRIS Reportable. There were no releases of radioactive materials, radiation exposures or personnel injuries associated with this event.